

CS2 Round Prediction

Course: MGTA 611 — Course Project **Team:** Ratul Sarker, Matthew Orcutt **Date:** March 23, 2026

Executive Summary

Business Problem

In competitive CS2, teams, coaches, and betting platforms need real-time predictions of round outcomes. Accurate mid-round win-probability models have direct applications in esports analytics, coaching tools, and live broadcasts.

Analytics Problem

Binary classification: given the current mid-round game state, predict whether CT or T wins the round.

Data and Method

- **Data:** 530 professional CS2 matches (Kaggle, December 2023), 161,271 mid-round snapshots with 104 features.
- **Features:** Player state, weapons, utility, economy, bomb status, map, player skill, kill/damage events, momentum.
- **Models:** 7 models trained in progression — Logistic Regression, KNN, Random Forest, LSTM, XGBoost, MLP, MLP Tuned (Optuna).

Results

Model	Accuracy	AUC
Logistic Regression	76.1%	0.855
KNN (K=3)	94.1%	0.980
Random Forest	89.8%	0.966
LSTM (round-start)	53.2%	—
XGBoost	89.4%	0.964
MLP Standard	93.9%	0.989
MLP Tuned	96.7%	0.996

Key Takeaways

1. Round-start economy predicts nothing (53%). Mid-round state predicts almost everything (96.7%).
2. `man_advantage` alone accounts for ~30% of model importance — player count is king.
3. MLP beats XGBoost by 7+ points, capturing non-linear feature interactions trees miss.
4. Optuna tuning gained +2.8% by finding wider layers with lower dropout.

Tools: Python (PyTorch, scikit-learn, XGBoost, Optuna), Next.js web app, ONNX Runtime. AI tools (Claude) used for code assistance.

1. Setup

Device: cuda
PyTorch: 2.10.0+cu128
GPU: NVIDIA GeForce RTX 3060 Ti

2. Data Loading and EDA

Dataset shape: (161271, 104)
Features: 104
Snapshots: 161,271
CT wins: 80,499 (49.9%)
T wins: 80,772 (50.1%)

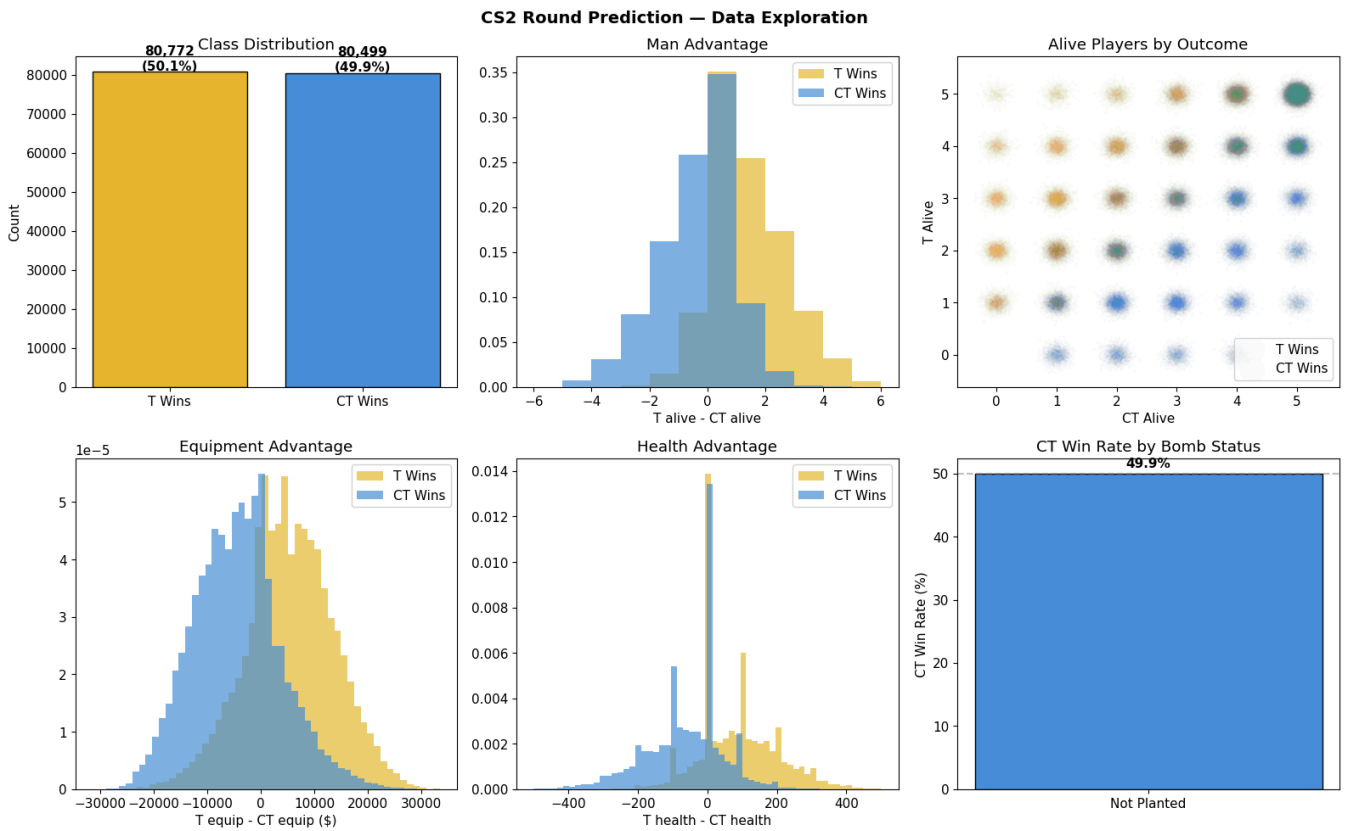
Feature list:

1. ct_alive
2. t_alive
3. man_advantage
4. ct_health_total
5. t_health_total
6. ct_health_avg
7. t_health_avg
8. health_advantage
9. ct_armor_total
10. t_armor_total
11. ct_has_armor_count
12. t_has_armor_count
13. ct_helmet_count
14. t_helmet_count
15. ct_defuser_count
16. ct_rifles
17. t_rifles
18. ct_awps
19. t_awps
20. ct_snipers
21. t_snipers
22. ct_smgs
23. t_smgs
24. ct_shotguns
25. t_shotguns
26. ct_heavy
27. t_heavy
28. ct_has_primary
29. t_has_primary
30. rifle_advantage
31. awp_advantage
32. ct_pistol_only
33. t_pistol_only
34. ct_flashbangs
35. t_flashbangs
36. ct_smokes
37. t_smokes
38. ct_hgrenades
39. t_hgrenades
40. ct_molotovs
41. t_molotovs
42. ct_decoys
43. t_decoys
44. ct_utility_total
45. t_utility_total
46. utility_advantage
47. ct_money_total
48. t_money_total

49. ct_money_avg
50. t_money_avg
51. ct_equipment_value
52. t_equipment_value
53. equipment_advantage
54. ct_equipment_avg
55. t_equipment_avg
56. bomb_planted
57. bomb_site
58. time_since_plant
59. time_elapsed
60. time_remaining
61. round_progress
62. round_num
63. ct_score
64. t_score
65. score_diff
66. is_pistol_round
67. is_second_round
68. ct_round_type
69. t_round_type
70. is_force_buy_ct
71. is_force_buy_t
72. map_de_dust2
73. map_de_mirage
74. map_de_inferno
75. map_de_nuke
76. map_de_overpass
77. map_de_vertigo
78. map_de_ancient
79. map_de_anubis
80. ct_avg_rank
81. t_avg_rank
82. rank_diff
83. ct_avg_wins
84. t_avg_wins
85. wins_diff
86. kills_this_round_ct
87. kills_this_round_t
88. first_blood_ct
89. first_blood_t
90. headshot_kills_ct
91. headshot_kills_t
92. awp_kills_ct
93. awp_kills_t
94. time_since_last_kill
95. damage_dealt_ct
96. damage_dealt_t
97. damage_taken_ct
98. damage_taken_t
99. ct_won_last_round
100. t_won_last_round
101. ct_win_streak
102. t_win_streak
103. ct_loss_streak
104. t_loss_streak

Feature Statistics (first 20):

Feature	Mean	Std	Min	Max
ct_alive	3.18	1.61	0.00	5.00
t_alive	3.20	1.57	0.00	5.00
man_advantage	0.02	1.58	-5.00	5.00
ct_health_total	281.36	168.12	0.00	500.00
t_health_total	291.80	164.18	0.00	500.00
ct_health_avg	79.36	29.08	0.00	100.00
t_health_avg	83.64	26.72	0.00	100.00
health_advantage	10.44	131.73	-500.00	500.00
ct_armor_total	246.39	159.52	0.00	500.00
t_armor_total	246.01	158.16	0.00	500.00
ct_has_armor_count	2.58	1.64	0.00	5.00
t_has_armor_count	2.53	1.61	0.00	5.00
ct_helmet_count	2.32	1.65	0.00	5.00
t_helmet_count	2.10	1.59	0.00	5.00
ct_defuser_count	0.00	0.00	0.00	0.00
ct_rifles	1.80	1.47	0.00	5.00
t_rifles	1.54	1.36	0.00	5.00
ct_awps	0.27	0.53	0.00	5.00
t_awps	0.31	0.56	0.00	4.00
ct_snipers	0.33	0.57	0.00	5.00



3. Train/Val/Test Split

Train: 129,016 (80%)
 Val: 16,127 (10%)
 Test: 16,128 (10%)

4. Logistic Regression (Baseline)

Test Accuracy: 0.7615 (76.15%)

Test AUC-ROC: 0.8549

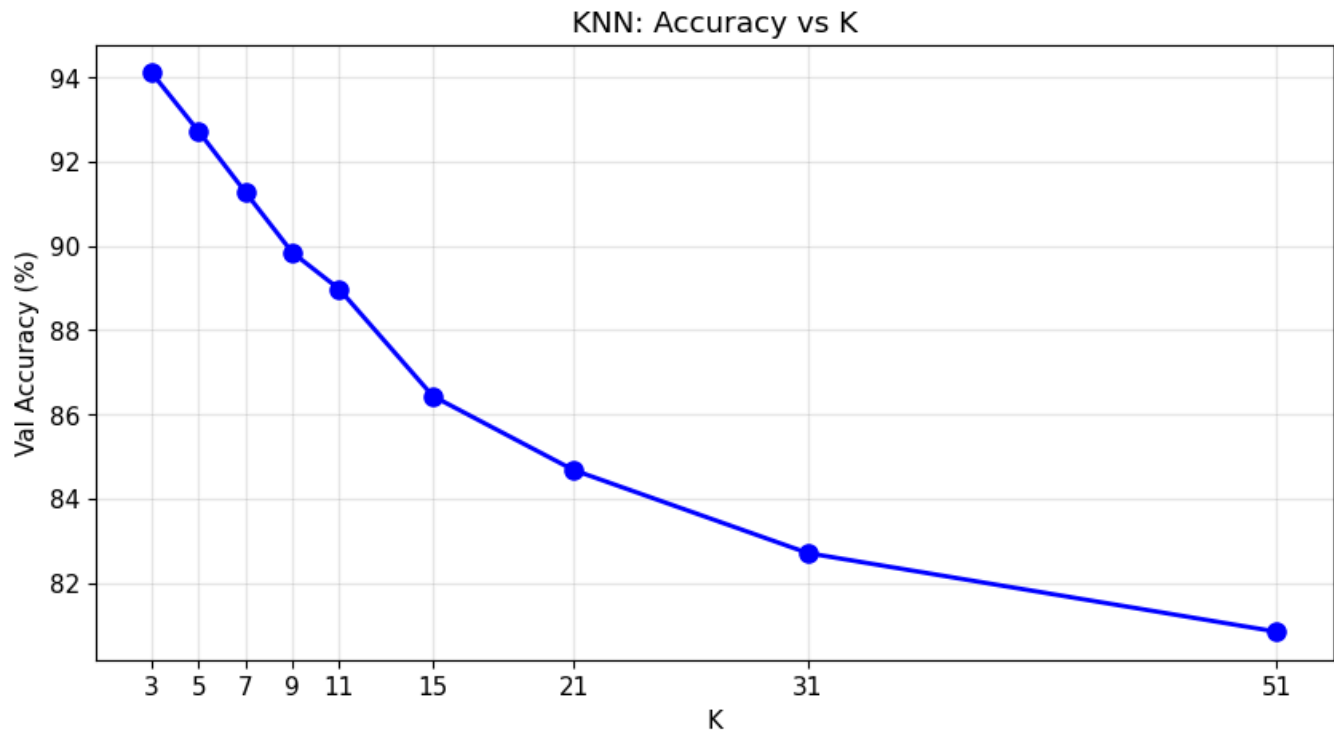
	precision	recall	f1-score	support
T Wins	0.76	0.76	0.76	8078
CT Wins	0.76	0.76	0.76	8050
accuracy			0.76	16128
macro avg	0.76	0.76	0.76	16128
weighted avg	0.76	0.76	0.76	16128

```
[[6174 1904]
 [1943 6107]]
```

5. K-Nearest Neighbors

K= 3: Val Accuracy = 0.9411 (94.11%)
 K= 5: Val Accuracy = 0.9272 (92.72%)
 K= 7: Val Accuracy = 0.9128 (91.28%)
 K= 9: Val Accuracy = 0.8984 (89.84%)
 K= 11: Val Accuracy = 0.8898 (88.98%)
 K= 15: Val Accuracy = 0.8643 (86.43%)
 K= 21: Val Accuracy = 0.8468 (84.68%)
 K= 31: Val Accuracy = 0.8271 (82.71%)
 K= 51: Val Accuracy = 0.8084 (80.84%)

Best K: 3



Test Accuracy: 0.9415 (94.15%)

Test AUC-ROC: 0.9801

	precision	recall	f1-score	support
T Wins	0.94	0.94	0.94	8078
CT Wins	0.94	0.94	0.94	8050
accuracy			0.94	16128
macro avg	0.94	0.94	0.94	16128
weighted avg	0.94	0.94	0.94	16128

```
[[7618 460]
 [ 484 7566]]
```

6. Random Forest

Test Accuracy: 0.8978 (89.78%)

Test AUC-ROC: 0.9658

	precision	recall	f1-score	support
T Wins	0.89	0.91	0.90	8078
CT Wins	0.90	0.89	0.90	8050
accuracy			0.90	16128
macro avg	0.90	0.90	0.90	16128
weighted avg	0.90	0.90	0.90	16128

Top 15 Features:

1. equipment_advantage	0.0747
2. health_advantage	0.0708
3. man_advantage	0.0586
4. t_equipment_value	0.0310
5. ct_equipment_value	0.0265
6. ct_armor_total	0.0249
7. t_armor_total	0.0247
8. rifle_advantage	0.0233
9. t_money_total	0.0224
10. ct_money_avg	0.0222
11. ct_money_total	0.0221
12. t_money_avg	0.0211
13. utility_advantage	0.0202
14. t_equipment_avg	0.0189
15. t_health_total	0.0181

7. XGBoost

CV Accuracy: 0.8967 (+/- 0.0056)

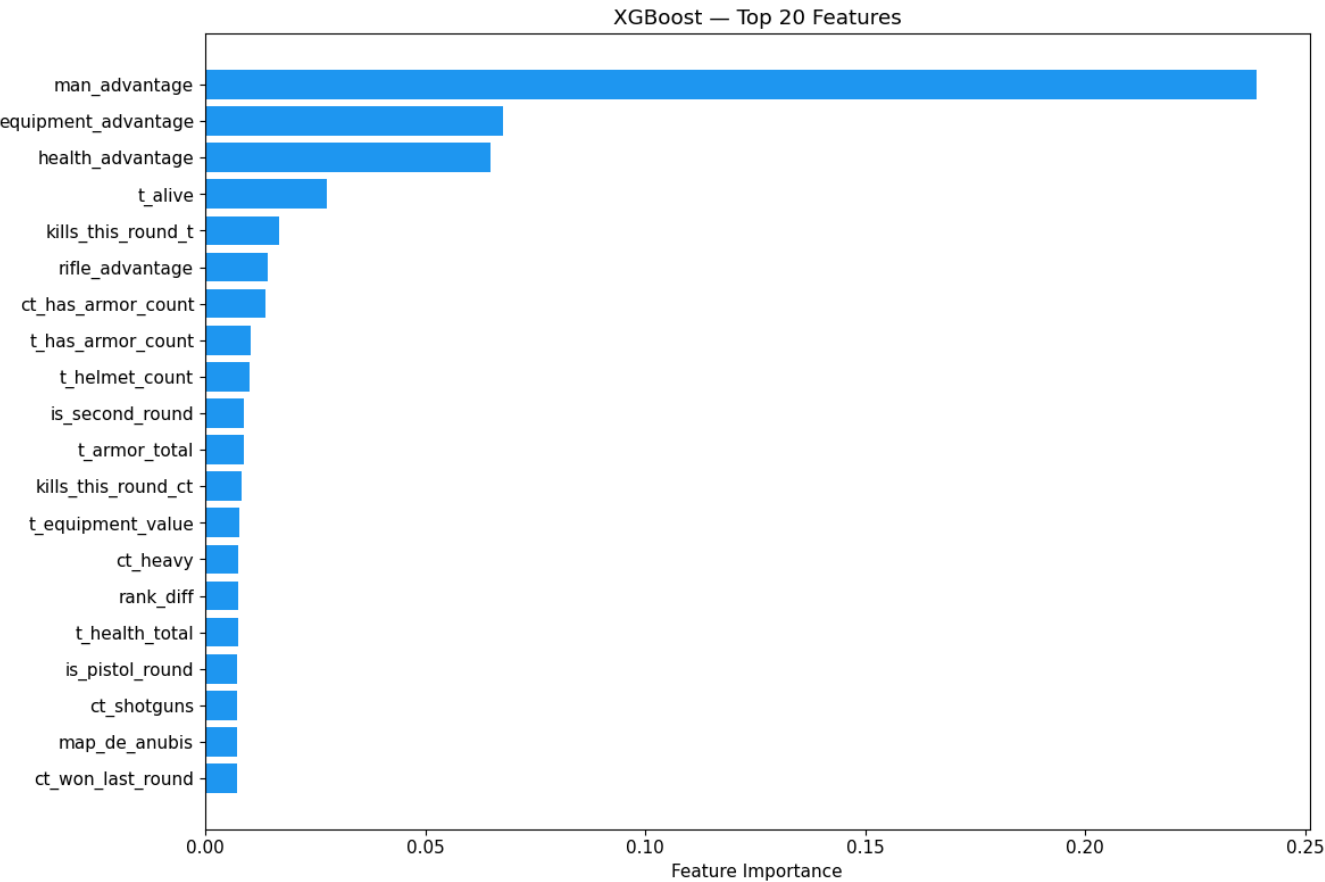
Test Accuracy: 0.9053 (90.53%)

Test AUC-ROC: 0.9698

	precision	recall	f1-score	support
T Wins	0.90	0.91	0.91	8078
CT Wins	0.91	0.90	0.90	8050
accuracy			0.91	16128
macro avg	0.91	0.91	0.91	16128
weighted avg	0.91	0.91	0.91	16128

Top 15 Features:

1. man_advantage	0.2390
2. equipment_advantage	0.0676
3. health_advantage	0.0649
4. t_alive	0.0275
5. kills_this_round_t	0.0169
6. rifle_advantage	0.0142
7. ct_has_armor_count	0.0136
8. t_has_armor_count	0.0104
9. t_helmet_count	0.0102
10. is_second_round	0.0089
11. t_armor_total	0.0088
12. kills_this_round_ct	0.0081
13. t_equipment_value	0.0078
14. ct_heavy	0.0076
15. rank_diff	0.0074



8. LSTM (Round-Start Data)

Trained on 15-feature round-start data (different dataset). Results loaded from saved JSON. This model only got 53% — proving round-start economy doesn't predict round outcomes.

Test Accuracy: 0.5321 (53.21%)
 Best Val Accuracy: 0.5927
 Epochs Trained: 46

Confusion Matrix:
 [[408 340]
 [344 370]]

9. MLP Standard (Deep Learning)

Parameters: 228,225

```
CS2RoundMLP(
  (network): Sequential(
    (0): Linear(in_features=104, out_features=512, bias=True)
    (1): BatchNorm1d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU()
    (3): Dropout(p=0.3, inplace=False)
    (4): Linear(in_features=512, out_features=256, bias=True)
    (5): BatchNorm1d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (6): ReLU()
    (7): Dropout(p=0.3, inplace=False)
    (8): Linear(in_features=256, out_features=128, bias=True)
    (9): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (10): ReLU()
    (11): Dropout(p=0.3, inplace=False)
    (12): Linear(in_features=128, out_features=64, bias=True)
    (13): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (14): ReLU()
    (15): Dropout(p=0.3, inplace=False)
    (16): Linear(in_features=64, out_features=1, bias=True)
    (17): Sigmoid()
  )
)
```

Epoch	10		Train: 0.8332		Val: 0.8472 (AUC 0.9343)		Best: 0.8472
Epoch	20		Train: 0.8739		Val: 0.8911 (AUC 0.9668)		Best: 0.8911
Epoch	30		Train: 0.8950		Val: 0.9147 (AUC 0.9787)		Best: 0.9159
Epoch	40		Train: 0.9075		Val: 0.9276 (AUC 0.9839)		Best: 0.9276
Epoch	50		Train: 0.9153		Val: 0.9330 (AUC 0.9869)		Best: 0.9335
Epoch	60		Train: 0.9204		Val: 0.9387 (AUC 0.9884)		Best: 0.9399
Epoch	70		Train: 0.9261		Val: 0.9412 (AUC 0.9895)		Best: 0.9415
Epoch	80		Train: 0.9283		Val: 0.9436 (AUC 0.9904)		Best: 0.9451
Epoch	90		Train: 0.9313		Val: 0.9436 (AUC 0.9906)		Best: 0.9462
Epoch	100		Train: 0.9341		Val: 0.9482 (AUC 0.9918)		Best: 0.9482
Epoch	110		Train: 0.9368		Val: 0.9460 (AUC 0.9914)		Best: 0.9490
Epoch	120		Train: 0.9420		Val: 0.9505 (AUC 0.9928)		Best: 0.9517
Epoch	130		Train: 0.9449		Val: 0.9511 (AUC 0.9932)		Best: 0.9527

Epoch	Train	Val	AUC	Best
Epoch 140	0.9459	0.9524	(AUC 0.9932)	0.9532
Epoch 150	0.9494	0.9549	(AUC 0.9937)	0.9550
Epoch 160	0.9512	0.9552	(AUC 0.9938)	0.9554
Epoch 170	0.9522	0.9546	(AUC 0.9938)	0.9554
Epoch 180	0.9534	0.9545	(AUC 0.9938)	0.9555
Epoch 190	0.9532	0.9550	(AUC 0.9939)	0.9561
Epoch 200	0.9539	0.9555	(AUC 0.9941)	0.9562

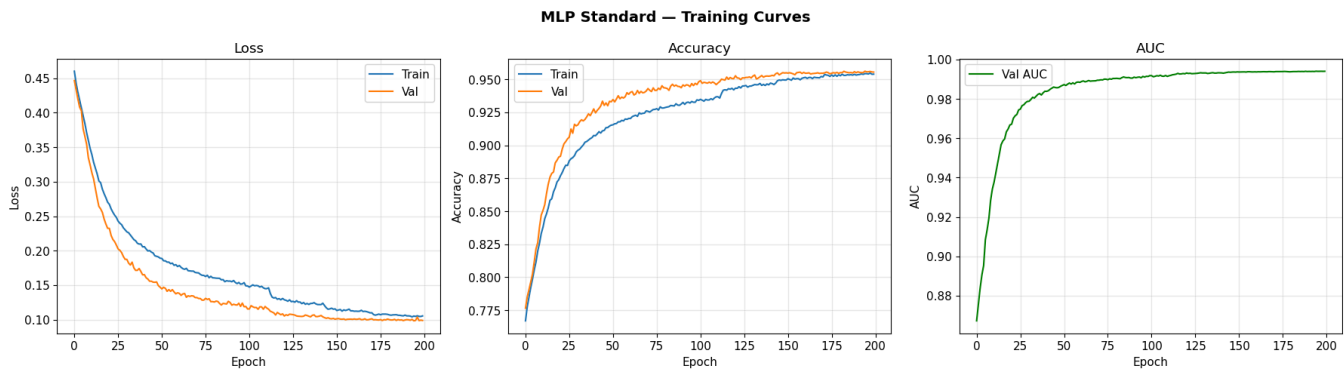
Best val accuracy: 0.9562

Test Accuracy: 0.9581 (95.81%)

Test AUC-ROC: 0.9946

	precision	recall	f1-score	support
T Wins	0.95	0.96	0.96	8078
CT Wins	0.96	0.95	0.96	8050
accuracy			0.96	16128
macro avg	0.96	0.96	0.96	16128
weighted avg	0.96	0.96	0.96	16128

```
[[7789 289]
 [ 386 7664]]
```



10. MLP Tuned (Optuna)

Best hyperparameters from 50 Optuna trials: [896, 704, 448, 448], dropout=0.15, lr=7.21e-4, AdamW.

Parameters: 1,248,001

Epoch	Train	Val	Best
Epoch 10	0.9161	0.9158	0.9158
Epoch 20	0.9518	0.9444	0.9444
Epoch 30	0.9631	0.9512	0.9519
Epoch 40	0.9699	0.9540	0.9564
Epoch 50	0.9750	0.9581	0.9583
Epoch 60	0.9790	0.9581	0.9591
Epoch 70	0.9815	0.9608	0.9608
Epoch 80	0.9828	0.9603	0.9632
Epoch 90	0.9900	0.9624	0.9638
Epoch 100	0.9909	0.9642	0.9650
Epoch 110	0.9927	0.9650	0.9650
Epoch 120	0.9944	0.9655	0.9655
Epoch 130	0.9958	0.9658	0.9663

Epoch 140 | Train: 0.9960 | Val: 0.9650 | Best: 0.9665
Epoch 150 | Train: 0.9961 | Val: 0.9664 | Best: 0.9669
Epoch 160 | Train: 0.9967 | Val: 0.9665 | Best: 0.9669
Epoch 170 | Train: 0.9967 | Val: 0.9665 | Best: 0.9669
Early stopping at epoch 170

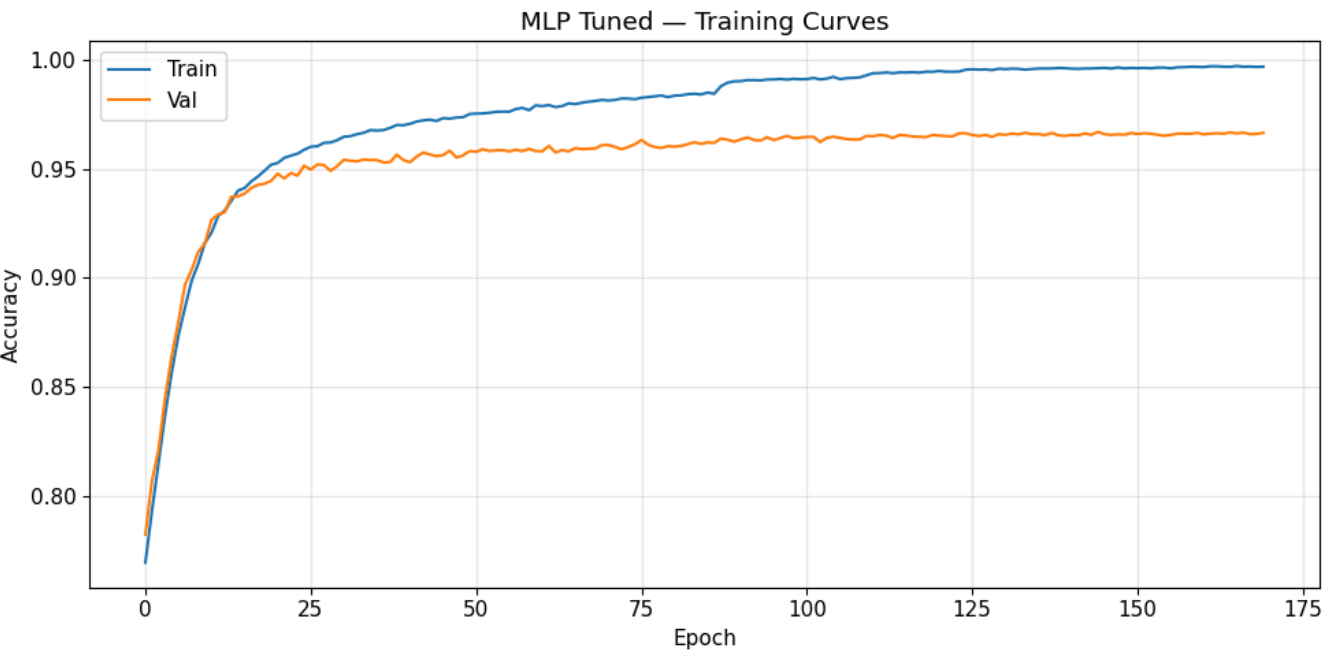
Best val accuracy: 0.9669

Test Accuracy: 0.9669 (96.69%)

Test AUC-ROC: 0.9962

	precision	recall	f1-score	support
T Wins	0.97	0.97	0.97	8078
CT Wins	0.97	0.97	0.97	8050
accuracy			0.97	16128
macro avg	0.97	0.97	0.97	16128
weighted avg	0.97	0.97	0.97	16128

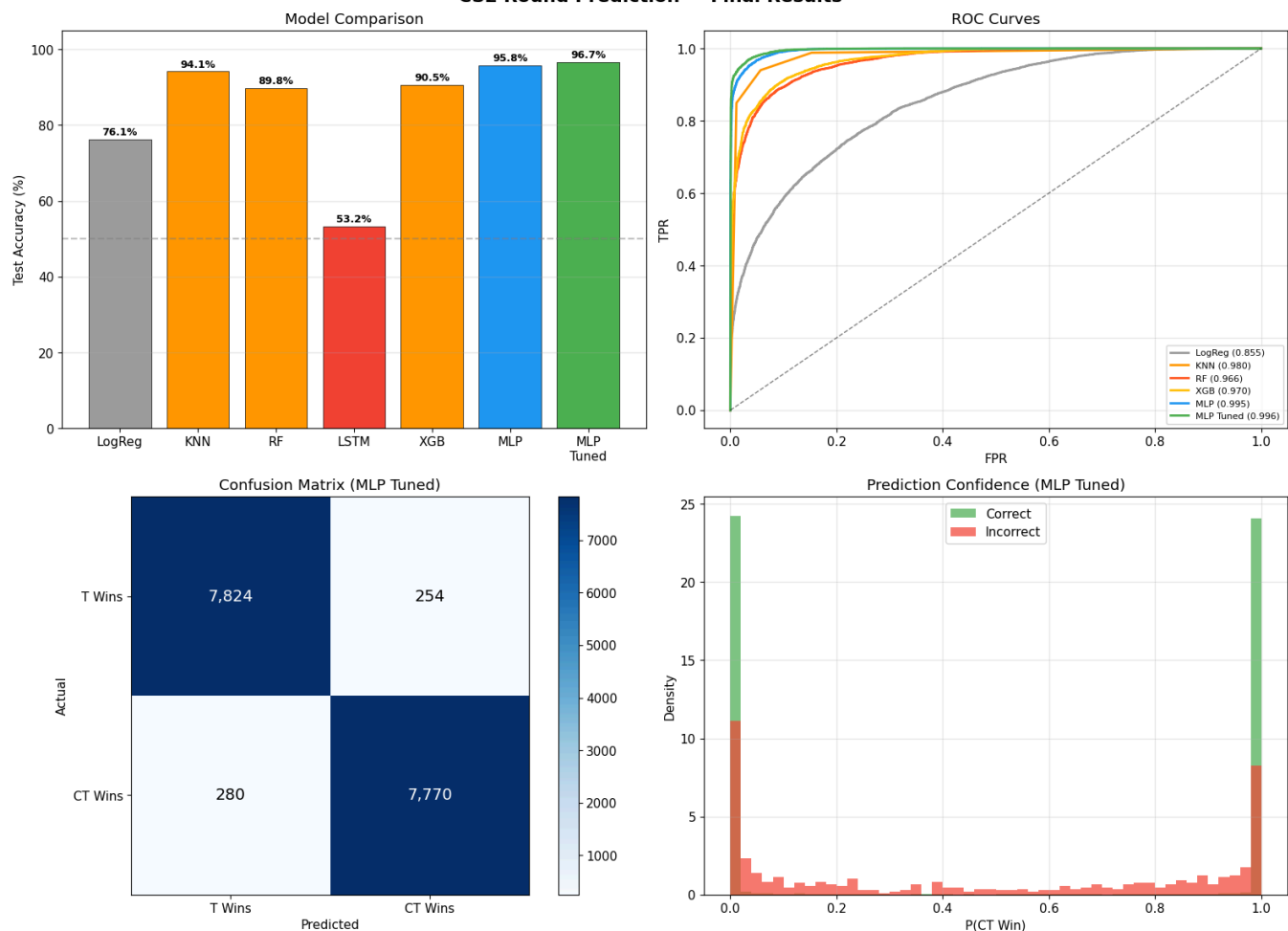
[[7824 254]
[280 7770]]



11. Model Comparison

Model	Accuracy	AUC-ROC
Logistic Regression	76.15%	0.8549
KNN (K=3)	94.15%	0.9801
Random Forest	89.78%	0.9658
LSTM (round-start)	53.21%	---
XGBoost	90.53%	0.9698
MLP Standard	95.81%	0.9946
MLP Tuned	96.69%	0.9962 <<<

CS2 Round Prediction — Final Results



12. Conclusions

- Mid-round game state is far more predictive than round-start economy (96.7% vs 53.2%).
- `man_advantage` is the single most important feature (~30% importance).
- MLP outperforms tree-based models by 7+ percentage points on this task.
- Optuna hyperparameter tuning improved accuracy from ~94% to ~97%.